

Trajectory of the Field

Trajectory: quỹ đạo or lộ trình

The future of data careers

AI

- Computer systems able to perform tasks that normally require human intelligence

Activities covered by the data professions

- Statistical inference
- Machine learning
- Data analytics

Current and future tools

One of the greatest skills a data professional can have is learning how to apply their knowledge of one tool to another tool. Throughout your career, you might discover that different organizations you work for use different tools—and in the field of data science, new and emerging technologies mean that exciting new tools are being developed all the time. This means there will always be opportunities to expand your data science toolkit! In this reading, you will learn more about tools today, including some of the tools you're going to learn about in this program. You will also explore some of the exciting ways tools are evolving and what that might mean for your toolkit in the future. Finally, you will explore a demonstration that illustrates how you'll be using some of these tools in the very near future.

Tools today

In this certificate program, you will have the opportunity to learn about many common tools data professionals use every day: spreadsheets, databases, query languages, data visualization, programming languages, and dashboards. Understanding the current tool landscape—and how it's changing—will help

you continue to grow your data science skills throughout your career. And understanding how the skills you learn for one tool can be applied to another means that you can adapt and add more tools to your toolkit!

Tool	Definition	Examples	Transferable skills
Spreadsheets	A digital worksheet where data can be manipulated and used for calculations	• Google Sheets • Microsoft Excel	• Data entry • Mathematical calculations • Manage datasets • Task automation • Data manipulation • Data analysis
Databases	A collection of data stored in a computer system	• Google Cloud • CloudSQL • Oracle • Microsoft SQL Server	• Database design • Data storage management • Data integrity
Programming languages	A system of words and symbols used to write instructions that computers follow	• SQL • R • Python • Java • C++	• Communicate with computer systems • Write and input commands • Manage datasets • Data manipulation • Data analysis
Data visualization	The graphical representation of data	• Tableau • Matplotlib • Seaborn • Google Charts • InfoGram • ChartBlocks	• Communicate data insights • Design compelling visuals • Identify key metrics
Dashboards	A tool that monitors live, incoming data	• Tableau • LookerStudio • Microsoft PowerBI	• Communicate data insights • Monitor real-time data • Develop data visualizations • Design filters and custom calculations

Already, there are so many tools to choose from as a data professional. This certificate program will focus primarily on Python and data visualizations. As

you progress in your career, you might find yourself learning new tools, and using your existing skills in new ways. Being able to recognize where tool skills overlap will help you continuously grow your data toolkit now and in the future.

You in the near future

So far in this reading, you have been considering how the skills you're going to learn in this certificate program will help you use even more tools in the future. As you prepare for your learning journey, you can also think about how you'll be able to apply these skills soon—not just in the distant future.

This certificate program focuses on some of the most commonly used tools for data analytics and machine learning with Python. More specifically, you will use:

- NumPy and pandas for data processing and manipulation
- matplotlib.pyplot, seaborn, and Tableau for visualizations
- statsmodels for statistical tests and regression modeling
- scikit-learn for building machine learning models

Next, consider the following overview of some of the tools you'll use to complete tasks in this certificate program.

You'll use pandas to view and manipulate tabular data with Python. In the following example, a comma-separated value (.csv) file is read into a pandas dataframe, of which the first five rows are displayed. A dataframe is basically a table used to organize data. This data is from the [UC Irvine Machine Learning Repository](#). It contains the count of public bicycles rented per hour in the Seoul Bike Sharing System, with corresponding weather data and holiday information.



You'll use NumPy and pandas to perform calculations and get statistics for your data.



You'll use Tableau, matplotlib.pyplot, and seaborn to create data visualizations.





You'll use statsmodels to practice statistical analysis and linear regression.



And you'll use scikit-learn to build and analyze machine learning models:



This is just a small sample of the full range of topics you'll learn about in this certificate. As you gain proficiency with these tools, you'll be equipped to take on nearly any data task.

Tools tomorrow and beyond

The world of data science is always growing and evolving—tools you might not have even known about a few years ago can quickly become necessary for professionals working in the field. As you consider the skills you are developing now, it can be useful to consider all the ways you might also use them in the future.

Artificial intelligence

Artificial intelligence, or AI, refers to computer systems that are able to perform tasks that normally require human intelligence. One of the great benefits of using AI for data science is that it can help provide real-time insights to stakeholders. For example, a business with an e-commerce website might use AI to monitor and deliver insights about how customers use their website in real-time, allowing the team to make quick improvements.

In today's dynamic workplace where leveraging the latest technology is key for productivity and efficiency, having an understanding of AI will boost your career as a data professional. While building a career in this industry, you can start enhancing your AI skills by exploring available AI tools that can assist with your role. One tool that's currently used is Tableau AI. Tableau AI aims to simplify the process of data analysis. This tool has the potential to help data professionals prepare data, reduce repetitive tasks, and suggest appropriate visualizations.

Machine learning

Machine learning is the use and development of algorithms and statistical models to teach computer systems to analyze and discover patterns in data. Data analysts can train algorithms to analyze large amounts of data that would otherwise take a long time to process. For example, a financial analyst might use machine learning to find patterns in the data that help identify fraud.

One of the most exciting developments in these future technologies is the way they can be used together to automate tasks and provide insights faster than ever.

Key takeaways

As a data professional, you will continue learning new skills and applying your current skills in new ways throughout your career. Recognizing how skills can be transferable allows you to adapt to different organizations' needs and evolving technologies. And as you progress through this, you add tools to your data science toolbox that will help you now and in the future!

How data professionals use AI

Earlier, you briefly considered the role of artificial intelligence in data science. You may recall that **artificial intelligence (AI)** refers to the development of computer systems able to perform tasks that normally require human intelligence. For example, practical applications of AI include voice assistants, autonomous vehicles, automated recommendation systems, and more.

In this reading, you will learn more about the uses of AI for data work, and how AI can help data professionals better understand their data and make more informed decisions. You'll also learn about the limitations of AI, and the differences between AI and human data professionals.

The uses of AI for data work

Data professionals can use AI to improve their data analysis, perform essential tasks, and streamline their workflow. For example, data professionals can use AI to:

- Create predictive models to help accurately forecast future events or outcomes.
- Automate time-consuming tasks such as data cleaning, coding, and report writing.

- Analyze extremely large datasets.
- Improve the quality of data by identifying and correcting errors.
- Generate insights from data that would not be obvious to humans.
- Provide guidance on tasks such as choosing the right algorithms and interpreting results.
- Facilitate collaboration among team members.

Data professionals can leverage AI to enhance the quality and efficiency of their data projects, generate valuable insights, and help stakeholders make better business decisions.

Conversational AI tools: Gemini and ChatGPT

Many data professionals now use conversational AI tools to help them analyze their data and boost their productivity. Two of the most frequently used tools are Gemini and ChatGPT. Gemini was created by Google AI. ChatGPT, also known as Chat Generative Pre-trained Transformer, was developed by OpenAI.

Gemini and ChatGPT are both **large language models (LLMs)** that are trained on massive datasets of text and code. An LLM is a type of AI algorithm that uses deep learning techniques to identify patterns in text and map how different words and phrases relate to each other. This allows LLMs to predict what word should come next. LLMs can generate human-like text in response to a wide range of prompts and questions.

Note: This is a general introduction to LLMs. A detailed discussion of the development and computational logic of LLMs is beyond the scope of this course.

Tools like Gemini and ChatGPT can help data professionals in a variety of ways. A data professional might ask Gemini or ChatGPT to:

- Clean a dataset by removing missing values, outliers, and duplicate data.
- Create interactive data visualizations such as dashboards and heatmaps.
- Recommend a specific algorithm for a particular task based on the data professional's input.
- Create a shared document to facilitate a brainstorming session among a team of data professionals.

Note: This is a brief list of possible prompts. In another activity, you'll get a chance to further explore Gemini and discover its capabilities.



Use cases for AI

Data professionals across industries use AI to help analyze data and generate insights for stakeholders. Here are some examples of how data professionals use AI in specific sectors:

Finance

- Analyze financial transactions to help prevent fraud and protect customers' money.
- Analyze large datasets of financial data to help identify potential risks and make more informed decisions about investments.
- Analyze historical market data and current market conditions to help generate sound investment recommendations.

Retail

- Recommend products to customers based on their past purchase history and browsing behavior.
- Track customers' interactions with the retail website to help personalize the shopping experience.
- Analyze sales data and forecast future demand to help optimize the amount of product inventory and reduce costs.

Manufacturing

- Automate tasks such as welding, painting, and assembly to help improve efficiency.
- Analyze data from sensors and cameras to help identify defects in products before they are shipped to customers.
- Analyze data from production lines to help identify ways to produce more products at a lower cost.

AI and human data professionals

Data professionals use AI as a tool to help them understand data, make better decisions, and improve efficiency. Like all tools, AI has limitations. Human data professionals possess skills, abilities, and qualities that AI currently lacks. For example:

- *Intuition.* AI models are trained on data, and they can only make decisions based on the patterns they observe in the data. Humans can use their intuition and personal experience to make decisions that are not explicitly programmed into the AI model. For this reason, it's important to always verify a model's output before relying on it.
- *Deal with ambiguity.* AI models are good at solving problems that are well-defined and have clear parameters. However, humans can identify and understand complex problems that are not well-defined and have ambiguous parameters by considering key details offered in the context of the project.
- *Interpersonal communication.* AI models can generate reports and presentations, but they cannot communicate with stakeholders in the nuanced way that humans can. Humans can explain the results of their analysis to fit the needs of specific stakeholders, and use their emotional intelligence to address concerns.
- *Creativity.* AI models are good at following instructions, but they are not imaginative like humans. Humans can be creative in their approach to data analysis, and imagine new and innovative solutions to complex problems.
- *Critical thinking.* Humans can think critically about their data and identify potential biases and ethical issues. AI models are usually trained on real-world data that contains biases and are therefore likely to reflect those biases in model outputs.
- *Leadership.* Humans can be leaders, and they can motivate and inspire others. AI may have difficulty understanding the nuances of human emotion, motivation, and communication. This limits AI's ability to effectively run an organization.
- *Factuality.* Generative AI models are trained to output text based on patterns in language. Sometimes the model output may be very well-composed and as a result, seem reliable, but may not be factual. As noted above, it's important to always verify model output.

In the future, product and research teams may develop updates for AI that enlarge its current capabilities. However, human data professionals will continue to play an important role in data science by using their intuition, imagination, and unique experience to solve complex problems.

Key takeaways

Data professionals can use AI to help automate tasks, make predictions, generate insights, and communicate findings. They can leverage AI to be more productive in their work and more impactful in their organizations. Overall, AI is a powerful tool for data professionals but it is not without limitations. For this reason, human oversight and intervention is critical when working with AI and related tools.

The places you'll go...

By now you're familiar with many of the tools data professionals use to overcome different challenges in their work. However, as important as these technical skills are for your professional development, creating engaging presentations of your work is just as important. As you complete this certificate program, you'll create a portfolio of projects that you can showcase to employers, other learners, or to broader audiences. In this reading you'll find examples of the many ways other data professionals in the field have shared their work, so that you can learn what such presentations entail, get a sense of what appeals to you, and perhaps even inspire you in your own journey.

Tableau dashboards

Bill Yost is a data analyst. One way he leverages his data professional skills is by building interactive data visualization dashboards on Tableau. His [project on office supply store analytics](#) is a great example of the dynamic capabilities Tableau offers data professionals.



His dashboard incorporates several different types of interactive graphs, including a bar chart, a line chart, and a map. You'll learn how to create these in this certificate program.

Tableau also allows you to create a profile to showcase your work. You can also follow other users and add projects that interest you to a list of favorites.



Tableau is a great platform if you want to feature polished visualizations and demonstrate your ability to communicate results in an engaging and elegant way.

Notebooks on Kaggle

If you prefer to compose a more technical exposition of your work, Kaggle is a great place to do it. For example, consider [Jesse Mostipak's profile](#). Jesse is a data professional who uses Kaggle to contribute coding exemplars and datasets, participate in modeling competitions, and converse with others in the data community.

The following is an example of a notebook she submitted as a tutorial on using XGBoost in tidymodels, which is a collection of R packages for modeling and machine learning. You'll learn how to use XGBoost with Python later in this certificate program.



In addition to supporting your public presence as a data practitioner, publishing your work this way is a great opportunity to generate discussion of different approaches and methodologies with peers in the data community and to learn new skills.

GitHub repositories

Another way to feature your work is by using GitHub. GitHub is probably best known for being a [version control system](#) used by coders and developers worldwide to keep track of projects involving large codebases with many simultaneous contributors. GitHub also offers tools to create and customize user profiles and project repositories. Take [Ravin Kumar's GitHub page](#) for example.



Some advantages of GitHub include the ability to structure your projects into repositories, which are like folders that are accessible to other people via Git. Repositories can include Jupyter notebooks, computer programs, images and figures, and markdown exposition. You'll work extensively with Jupyter notebooks and markdown text throughout this certificate program. GitHub also lets you watch, star, and follow users or individual projects. The platform is particularly useful for sharing code directly with other people and even working collaboratively with them on projects.

Blogs on Medium

Jupyter notebooks and coding exhibitions aren't the only means by which you can demonstrate expertise. Blog posts are another great way to propel your professional development. One of the advantages of blogs is that they showcase a wide range of highly sought-after skills beyond just technical proficiency. For example, not only will a well-written blog post convey your understanding of data concepts, tools, and techniques, but it will also emphasize your critical thinking, communication, and writing skills. These types of skills are what distinguish the best data professionals in the field.

Medium is a great platform to post data science-related blogs. You can also submit to publishers of curated data science content, such as Towards Data Science, Analytics Vidhya, and KDNuggets. As an example, consider the following [article written by Cassie Kozykov](#) (the instructor of this course) that was published on Towards Data Science.



This post isn't about coding or complex computer science or mathematics at all! It's a practical guide for avoiding pitfalls when working with third-party data. Such topics are highly relevant, and writing about them is another great way to access exciting career opportunities.

Other approaches

This list isn't exhaustive. It's intended to give you an idea of the possibilities for presenting your work and becoming an active participant in the field. Some people share Jupyter notebooks, others write helpful articles, and others create podcasts or YouTube videos. For example, consider [this YouTube video on bias](#)

in machine learning and AI, produced by Angelica Spratley, a data scientist and educator.



By the way, bias is something you'll learn about in the course on machine learning. Find what excites you, and your enthusiasm will be evident. The possibilities are vast if you follow your imagination and motivation to contribute to the exciting world of data.

Key Takeaways

These are just a few ways you can share your work. One of the great things about doing so is that it gives you a public-facing profile that employers can review. If someone searches for your name and finds an impressive portfolio, it gives you more opportunity to stand out. Additionally, presenting your work forces you to clearly explain (and thus understand) your methodologies, making you a stronger data practitioner. Whichever way you choose to share your work, be creative, have fun, and don't be afraid to put yourself out there. This is your opportunity to impress!